

B. Tech Degree VIII Semester Examination April 2011

EC/EB/EI 804 (C) MECHATRONICS (2002 Scheme)

Time : 3 Hours

Maximum Marks : 100

- I. (a) What are the common criteria based on which NC machines are classified? Explain PTP, CP and straight cut types of NC machines with typical examples. (15)
(b) Explain the need of pre-loading ball screws in CNC machines. Explain the methods of pre-loading ball screws with neat sketches. (10)
- OR**
- II. (a) Explain the relevance of 'mechatronics' based on the development of the following products: (i) automatic washing machine (ii) thread mills. (10)
(b) What are the components of a CNC machine? Explain with the help of block diagrams/sketches. (15)
- III. (a) Sketch and explain the working of (i) Fluidic and (ii) electronic bistable flip flops (15)
(b) Explain the linear and circular interpolations with the help of a neat sketch (example) (10)
- OR**
- IV. (a) Discuss the working of DC servo motors and stepping motors. (15)
(b) What is an Inductosyn? Explain its working principle with the help of a neat figure. (10)
- V. What are the functional elements of CIM? Explain each with typical activities. Explain how the integration of the different functional elements are achieved in a typical CIM system. (25)
- OR**
- VI. (a) What is a "block" in part programming? Write a complete word address and explain each element. (15)
(b) Explain the following terms in programming (i) subroutines (ii) mirror image (iii) tool compensation. (10)
- VII. (a) Explain (i) Resolution (ii) Repeatability of robotic systems. (10)
(b) Explain the five basic configurations commonly available in commercial industrial robots. (15)
- OR**
- VIII. (a) Explain different robotic programming methods. (15)
(b) Explain Robot language structure. Also explain the generations of Robot programming languages. (10)